

On the reclassification of A2 in TECT: from an independent ultra-high-energy postulate to a high-temperature symmetric-phase boundary class

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We analyse the logical role of axiom A2 (an ultra-high-energy isotropic initial state at $t = -\infty$) in the foundations of topological energy condensate theory and show that A2 admits a *weakening / reclassification*, NOT an elimination. Given a Brazovskii free-energy functional A0, dissipative TDGL kinetics A1, and an externally supplied cosmological cooling history $T(t)$ with an early-time regime $T \gg T_c$, the original axiom A2 need not be imposed as an independent microscopic dynamical law. It can be replaced by the weaker requirement that the early-time state belongs to the symmetric high-temperature phase *class*, characterised at the coarse-grained level by $\langle \Psi \rangle = 0$ and full unbroken spatial / internal symmetry. Two structural caveats prevent a stronger “elimination” claim: (a) the dissipative TDGL flow is NOT time-reversible, so a unique microscopic backward extrapolation to $t = -\infty$ cannot be invoked; (b) the ensemble-level phase-class statement $\langle \Psi \rangle = 0$ does not single out a unique microscopic Cauchy datum. The total number of premises is preserved (3 = 2 internal axioms + 1 external boundary input); the reclassification moves $T(t)$ from an internal TECT axiom to an external cosmological input rather than eliminating it.

I. INTRODUCTION

In constructing a candidate theory-of-everything framework, a useful structural question is: how many genuinely independent internal assumptions does the theory require, and which of those assumptions are properly internal as opposed to being inherited from external boundary data? A theory with fewer genuinely independent internal assumptions is, all else equal, structurally more economical, in the sense that fewer postulates are subject to falsification at the level of the theory itself.

TECT originally posited three independent axioms (A0, A1, A2). The present work explores whether one of these—the ultra-high-energy initial state (A2)—can be derived from the other two in the presence of cosmological inputs. If so, the logical structure of TECT simplifies from a "3-axiom framework" to a "2-axiom + 1-boundary-condition framework," moving one axiom from the theory

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itself to the specification of the problem domain.

II. SETTING AND MAIN RESULT

A. The three TECT axioms

1. **A0 (Brazovskii functional):** The primordial BCC condensate is governed by a Brazovskii-type free-energy functional $\mathcal{F}[\Psi]$ with locked parameters (μ^2, λ, γ) and shell radius q_0 . This is the dynamical foundation: it specifies the equations of motion via TDGL.
2. **A1 (TDGL kinetics):** The time evolution is governed by time-dependent Ginzburg–Landau (TDGL) Model-A kinetics:

$$\partial_t \Psi = -\Gamma \frac{\delta \mathcal{F}}{\delta \Psi^*} + \xi(t), \quad (1)$$

where Γ is the Onsager relaxation coefficient and ξ is thermal noise (possibly suppressed in the zero-temperature limit).

3. **A2 (Ultra-high-energy isotropic initial state):** At $t = -\infty$ (or $t \rightarrow -\infty$), the order parameter is symmetric and isotropic, $\langle \Psi \rangle = 0$, with energy density set by the high-temperature regime $T \gg T_c$. Full $O(3) \rtimes \mathbb{R}^3 \times U(1)$ symmetry.

B. Main theorem

Theorem 1 (A2 weakening / reclassification). *Assume:*

1. *A Brazovskii-type free-energy functional $\mathcal{F}[\Psi; T]$ with locked parameters $(\mu^2, \lambda, \gamma, q_0)$ depending smoothly on temperature T (axiom A0).*
2. *Dissipative TDGL evolution toward lower free energy $\partial_t \Psi = -\Gamma \delta \mathcal{F} / \delta \Psi^*$ (axiom A1).*
3. *An externally supplied cosmological cooling history $T(t)$ with an early-time regime $T \gg T_c$ (boundary input, NOT an internal TECT axiom).*

Then the original axiom A2 (independent ultra-high-energy isotropic initial state) need NOT be imposed as an independent microscopic dynamical law. It can be replaced by the weaker requirement that the early-time state belongs to the symmetric high-temperature phase class, characterised at the coarse-grained level by

$$\langle \Psi \rangle = 0 \quad \text{and} \quad \text{full unbroken spatial / internal symmetry.}$$

The total premise count is preserved as $3 = 2_{\text{internal}} + 1_{\text{boundary}}$; A2 itself is removed from the internal-axiom list, and $T(t)$ is moved from an implicit internal assumption to an explicit external boundary input.

Remark 2 (Why this is weakening, not elimination). Two structural obstacles prevent a stronger “elimination” claim. (i) The dissipative TDGL flow $\partial_t \Psi = -\Gamma \delta \mathcal{F} / \delta \Psi^*$ is NOT time-reversible; a backward extrapolation from a late-time observation to a unique microscopic Cauchy datum at $t = -\infty$ is not available. (ii) The phase-class statement $\langle \Psi \rangle = 0$ is an ensemble / coarse-grained characterisation, not a single microscopic configuration; it admits many distinct microscopic Cauchy data compatible with the same phase-class behaviour. The theorem above therefore provides a *reclassification* of A2 (from an independent internal axiom to a phase-class boundary requirement consistent with A0 + A1 + $T(t)$), not a derivation of a unique microscopic initial state.

III. PROOF SKETCH

a. Step 1 (Cauchy-problem formulation). The TDGL dynamics defines a Cauchy problem: given an initial state $\Psi(t_0)$ at some time t_0 , the equation

$$\partial_t \Psi = -\Gamma \frac{\delta \mathcal{F}}{\delta \Psi^*} \quad (2)$$

(ignoring noise in the zero-temperature limit) uniquely determines $\Psi(t)$ for all $t > t_0$.

b. Step 2 (Forward consistency, NOT backward time-reversibility). The dissipative TDGL flow is NOT time-reversible (cf. Remark 2); we therefore do NOT attempt a microscopic backward extrapolation from a late-time observation. The relevant statement is forward consistency: under TDGL evolution, states starting in the symmetric high-temperature phase $\langle \Psi \rangle = 0$ at $t \rightarrow -\infty$ flow into the broken-phase BCC basin only after the temperature crosses T_c . Backward uniqueness of the microscopic Cauchy datum is NOT used.

c. Step 3 (High-temperature limit). At $t = -\infty$, the early universe was hot: $T \gg T_c$. In the high-temperature regime, the free energy $\mathcal{F}[\Psi]$ is dominated by the quadratic term $\sim \mu^2 |\Psi|^2$ with $\mu^2 > 0$ (symmetric phase). The lowest-energy state is $\Psi = 0$ (isotropic vacuum). This is not a postulate—it is a thermodynamic consequence of the free-energy form at high T .

As the universe cools ($T(t)$ decreases), the parameters $\mu^2(T)$ and $\lambda(T)$ evolve. At some critical temperature T_c , a phase transition occurs: $\mu^2 \rightarrow 0$ and the condensate forms.

d. Step 4 (Phase-class argument, NOT unique-Cauchy-datum argument). The statement reframed by the present theorem is: “the early-time state belongs to the symmetric high-temperature

phase class $\langle \Psi \rangle = 0$ ". This is a phase-class statement, compatible with many distinct microscopic Cauchy data; it is NOT the claim that the microscopic Cauchy datum is unique. The phase-class statement is the natural ensemble-level characterisation of the symmetric high-temperature regime $T \gg T_c$ given A0 + A1, and is *weaker* than the original A2 (which postulated a particular microscopic isotropic vacuum). The reclassification is therefore a genuine theoretical-economy gain (one fewer internal microscopic postulate) at the price of an explicit external boundary input $T(t)$.

e. Step 5 (Honest scope). Two irreducible elements remain:

1. The existence of a $t = -\infty$ boundary (fundamental to any Cauchy problem, not specific to TECT).
2. The cosmological temperature evolution $T(t)$ (input from general relativity / inflation, not derived from TECT alone).

These two elements constitute the "1 cosmological boundary condition" and cannot be absorbed into TECT without external input.

IV. DISCUSSION AND OUTLOOK

The reclassification result of Theorem 1 moves the ultra-high-energy state from an independent internal microscopic postulate to a phase-class boundary requirement consistent with the external cosmological input $T(t)$. The total premise count is *preserved* as

$$3 = 2_{\text{internal}} + 1_{\text{boundary}},$$

i.e. the change is *not* an axiom-count reduction; it is a relabelling of one of the three premises from an internal TECT axiom to an externally supplied cosmological input. The structural gain is clarity, not parsimony in the strict count.

a. What the result does NOT establish. Per Remark 2, the dissipative TDGL flow is not time-reversible and the phase-class statement $\langle \Psi \rangle = 0$ does not single out a unique microscopic Cauchy datum at $t = -\infty$. The result is therefore a *weakening* / *reclassification* of A2, not an *elimination*, and *not* a derivation of a unique microscopic initial state. TECT's internal logical structure still depends on three premises, of which two (A0 and A1) are internal dynamical postulates and one (an externally supplied $T(t)$) is a cosmological boundary input that is not derived inside TECT.

b. Future work. A genuine premise-count *reduction* would require deriving $T(t)$ from a TECT-internal cosmological mechanism (or from general relativity coupled to a TECT-internal stress–energy source), so that the externally supplied boundary input becomes a derived quantity. Pending such a derivation, the present reclassification result is the strongest internally-supported statement.

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